Galaxy Morphology Classification Under Reduced Image Resolution

Miko Samson
1009022111, miko.samson@mail.utoronto.ca
`https://github.com/miko-oh/aps360-project`

Introduction

Galaxy morphology describes the shape and structure of galaxies, including their spiral arms, bars, and bulges. Studying these features helps astronomers understand how galaxies form and evolve. Due to the sheer number of these celestial bodies, manual classification would be extremely time consuming and difficult, promoting the introduction of an automated method [Willett et al., 2013, Dieleman et al., 2015]. Deep learning is appropriate for this task because a model can learn visual features from labelled images and automate galaxy classification.

I will investigate how reduced image resolution affects the ability of a convolutional neural network (CNN) to distinguish ten galaxy classes. A CNN is suitable because its convolutional layers learn the spatial patterns and details directly from images. The project will compare its performance against a conventional classifier and identify the classes which are most sensitive to reduced resolutions.

Illustration

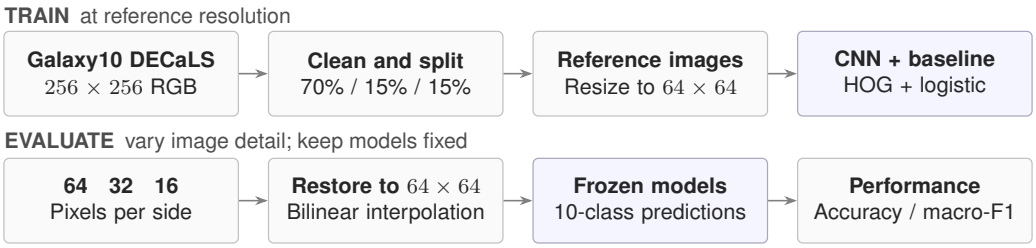


Figure 1: Both models are trained on the 64px reference images. Evaluation uses the same test images at three different detail levels, 32px and 16px inputs scaled up to 64px resolutions. At each resolution their performances are compared.

Background & Related Work

Willett et al. [2013], who released Galaxy Zoo 2 by taking volunteer classifications to describe galaxy morphology. Additionally, Hart et al. [2016] developed a method to correct redshift-dependent classification bias, correcting some observational effects on the dataset. Moreover, Dieleman et al. [2015] produced a CNN which exploited the rotational and translational symmetry for morphology prediction. Furthermore, Walmsley et al. [2022] combined volunteer labels with Bayesian CNNs for Galaxy Zoo Dark Energy Camera Legacy Survey (DECaLS) images and showed that deeper imaging revealed additional features which would aid in classification. Lastly, Pandya et al. [2023] investigated equivariant networks on Galaxy10 DECaLS and tested its robustness using noise and single-pixel changes. Building upon all of these studies, this project will examine the controlled loss of spatial detail using a compact CNN and conventional baseline for evaluation.

Data Processing

I will be using the original Galaxy10 DECaLS dataset, which contains 17,736 RGB 256px images of ten morphology classes [Leung and Bovy, 2024]. Before splitting the data for the training and evaluation process, the data is audited for constant images or potential duplicates. Rows that are connected by identical image hashes or sky positions within one arc-second are grouped. Groups with conflicting labels are excluded, otherwise, one nonconstant image is kept. Nearby coordinates act as a conservative proxy and can group distinct objects. The initial audit kept 17,615 images which were then split into their respective training, validation, and testing groups with a stratified 70/15/15 ratio, seed 42 for reproducibility.

All images are resized to a 64px reference size. CNN channel normalization statistics are calculated from training images only. When evaluating, the reference images are left unchanged or downsampled to 32px or 16px, then restored to 64px using bilinear interpolation. This process lets the model input dimensions be fixed while varying the detail available. This data will be used for both models.

Architecture

I will train a compact CNN from scratch using PyTorch. It will contain three blocks with 16, 32, and 64 channels, all using two 3×3 convolutions, ReLU activations, and max pooling, with batch normalization after the first convolution. This is to improve generalization and limit complexity. Adaptive average pooling and a fully connected layer with 128 units will lead to ten output logits, with dropout for regularization. For training, it will use 64×64 reference images, random 90-degree rotations and horizontal flips, class-weighted cross-entropy, and Adam. Validation macro-F1 will guide checkpoint selection and early stopping.

Once the model choices are determined, I will evaluate both models on the test set at all three detail levels. Accuracy, macro-F1, and confusion matrices will evaluate the performance of the model's classification and common errors. The analysis will compare performance loss with resolution and inspect class-specific failures.

Baseline Model

The baseline model will use a histogram of oriented gradients (HOG) features and each RGB channel's mean and standard deviation, followed by standardized features and class-balanced logistic regression. HOG will use nine orientation bins, the 8×8-pixel cells, and 2×2-cell blocks. The regularization parameter C is then selected from $\{0.1, 1, 10\}$ using validation macro-F1. Similar to the CNN, the model will also be trained on reference images and evaluated at all the three detail levels.

Ethical Considerations

While the galaxy images pose minimal privacy risk, and the dataset was labeled by volunteers, it is possible that there are some uncertainties, potentially introducing a bias to the scientific conclusions. These errors for any particular morphology class may propagate and distort estimates of galaxy populations. I will report class-level performance and document cleaning to promote transparency and express the limitations. The downsampling also approximates a particular aspect of the image quality, in which conclusions will be limited to the dataset and degradation procedure only. This classifier should support scientific review rather than being treated as ground truth.

References

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